MIND CONTROLLED ROBOTIC HAND

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1. INTODUCTION

RECENT ADVANCEMENTS IN ROBOTICS AND BRAIN-COMPUTER INTERFACES HAVE ENABLED DIRECT COMMUNICATION BETWEEN THE HUMAN BRAIN AND ELECTRONIC DEVICES. THIS PROJECT PRESENTS A ROBOTIC HAND THAT CAN BE CONTROLLED THROUGH BRAIN-GENERATED COMMANDS, REDUCING THE DEPENDENCE ON TRADITIONAL MANUAL CONTROLS.

THE ROBOTIC HAND IS INTENDED PRIMARILY FOR ASSISTING PEOPLE WITH DISABILITIES WHO HAVE LIMITED HAND MOVEMENT. BY COMBINING AN ARDUINO MICROCONTROLLER, SERVO MOTORS, AND THE UPSIDE LABS MIND-CONTROL TECHNOLOGY, THE PROJECT DEMONSTRATES HOW BRAIN SIGNALS CAN BE TRANSLATED INTO MECHANICAL MOVEMENT.

2. PROBLEM STATEMENT

PEOPLE WHO HAVE LOST THE USE OF THEIR HANDS DUE TO PARALYSIS, AMPUTATION, SPINAL CORD INJURIES, STROKE, OR NEUROMUSCULAR DISORDERS OFTEN STRUGGLE TO PERFORM BASIC DAILY TASKS INDEPENDENTLY. EXISTING PROSTHETIC HANDS AND ASSISTIVE ROBOTIC DEVICES CAN BE EXPENSIVE, DIFFICULT TO OPERATE, OR REQUIRE PHYSICAL MUSCLE MOVEMENT THAT SOME USERS CANNOT PRODUCE.

AS A RESULT, MANY INDIVIDUALS REMAIN DEPENDENT ON CAREGIVERS, REDUCING THEIR INDEPENDENCE AND QUALITY OF LIFE.

1.3 BILLION PEOPLE, OR ABOUT **16%** OF THE GLOBAL POPULATION, EXPERIENCE SIGNIFICANT DISABILITY. THIS MEANS ROUGHLY 1 IN 6 PEOPLE WORLDWIDE LIVE WITH SOME FORM OF DISABILITY.

ABOUT 80% OF DISABLED INDIVIDUALS LIVE IN DEVELOPING COUNTRIES.

AS PER THE OFFICIAL CENSUS 2011, THERE ARE **2.68 CRORE (26.8 MILLION)** DISABLED PEOPLE IN INDIA, MAKING UP **2.21 PERCENT** OF THE TOTAL POPULATION. OUT OF THIS TOTAL, ABOUT 1.50 CRORE ARE MALE AND 1.18 CRORE ARE FEMALE.

3. OBJECTIVES

- TO ACQUIRE EEG SIGNALS FROM THE HUMAN BRAIN USING SCALP ELECTRODES.

- TO AMPLIFY AND PROCESS THE EEG SIGNALS USING THE BIO AMP EXG PILL.

- TO INTERFACE THE EEG SIGNAL ACQUISITION SYSTEM WITH AN ARDUINO.

- TO CONTROL FOUR SERVO MOTORS BASED ON PROCESSED BRAIN SIGNALS.

Component	Quantity
Arduino	5
BioAmp EXG Pill	1
Disposable EEG Electrodes	3
Servo Motors	5
Robotic Hand Frame	1
Jumper Wires	As required
USB Cable	1
Power Supply	1










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4. WORKING PRINCIPLE

- THE THREE ELECTRODES ARE CONNECTED TO THE USERS HEAD TO CAPTURE EEG SIGNALS.

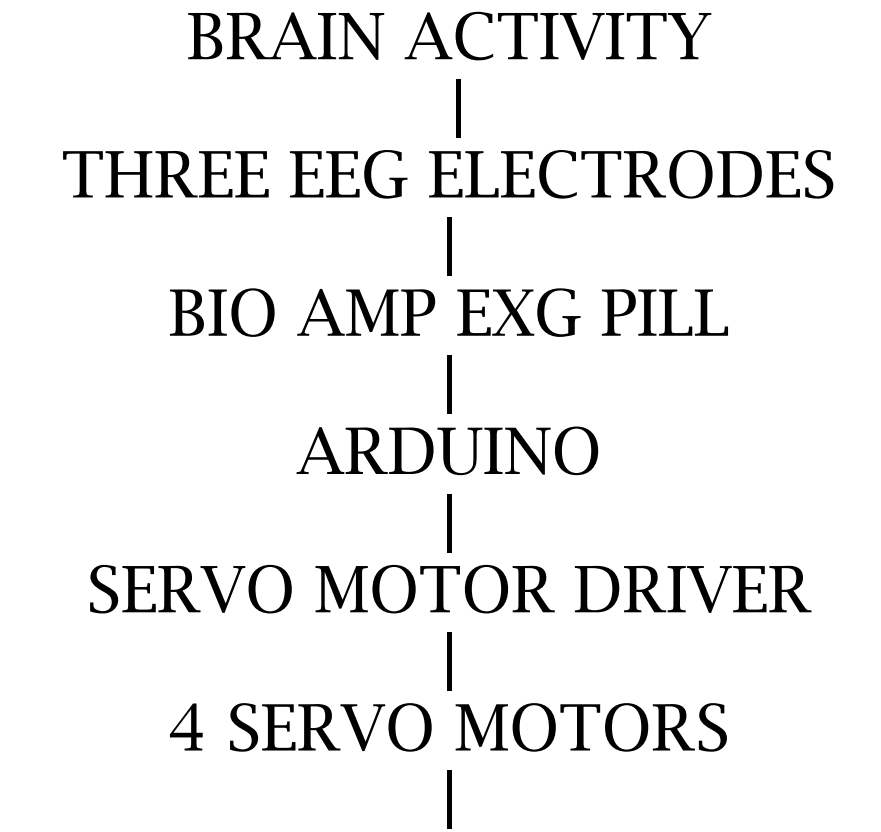
- THE BIO AMP EXG AMPLIFIES WEAK ELECTRICAL SIGNALS BY THE BRAIN.

- THE RECORDED SIGNALS ARE PROCESSED SHOWN IN THE ARDUINO

- THE WAVES ARE RECORDED AND SENT TO THE 5 SERVO MOTORS

- THE SERVO MOTORS MOVE THE FINGERS OF THE ROBOTIC HAND, PRODUCING ACTIONS SUCH AS OPENING AND CLOSING.

5. BLOCK DIAGRAM



6. HARDWARE DESCRIPTION

BIO AMP EXG PILL

THE BIO AMP EXG PILL IS A COMPACT BIO SIGNAL AMPLIFIER DESIGNED TO CAPTURE VERY WEAK BIOLOGICAL ELECTRICAL SIGNALS SUCH AS EEG, ECG, AND EMG. IN THIS PROJECT, IT AMPLIFIES EEG SIGNALS COLLECTED FROM THE SCALP BEFORE SENDING THEM TO THE ARDUINO.

EEG ELECTRODES

THREE ELECTRODES ARE PLACED ON THE HEAD TO DETECT ELECTRICAL ACTIVITY PRODUCED BY THE BRAIN. PROPER PLACEMENT ENSURES STABLE SIGNAL ACQUISITION.

ARDUINO

THE ARDUINO ACTS AS THE MAIN CONTROLLER. IT RECEIVES THE AMPLIFIED EEG SIGNALS, PROCESSES THEM, AND GENERATES PWM OUTPUTS FOR CONTROLLING THE SERVO MOTORS.

SERVO MOTORS

FOUR SERVO MOTORS ARE ATTACHED TO THE ROBOTIC FINGERS. EACH MOTOR ROTATES TO A SPECIFIED ANGLE BASED ON THE CONTROL SIGNAL FROM THE ARDUINO, ENABLING FINGER MOVEMENT.

7. METHODOLOGY

- PLACE THREE EEG ELECTRODES ON THE SCALP.

- CONNECT THE ELECTRODES TO THE BIO AMP EXG PILL.

- CONNECT THE BIO AMP EXG PILL OUTPUT TO THE ARDUINO ANALOG INPUT.

- UPLOAD THE ARDUINO PROGRAM FOR SIGNAL PROCESSING.

- CONNECT THE FOUR SERVO MOTORS TO THE ARDUINO PWM PINS.

- MONITOR EEG SIGNALS.

- WHEN THE REQUIRED EEG SIGNAL PATTERN IS DETECTED, THE ARDUINO COMMANDS THE SERVO MOTORS TO MOVE THE ROBOTIC HAND.

8. APPLICATIONS

- ASSISTIVE TECHNOLOGY FOR PEOPLE WITH PARALYSIS.
- PROSTHETIC HAND CONTROL.
- REHABILITATION THERAPY.
- BRAIN-COMPUTER INTERFACE RESEARCH.
- HUMAN-MACHINE INTERACTION.
- MEDICAL AND HEALTHCARE ROBOTICS.
- EDUCATIONAL AND RESEARCH PROJECTS.

9. ADVANTAGES

- LOW-COST IMPLEMENTATION.
- NON-INVASIVE BRAIN SIGNAL ACQUISITION.
- EASY TO PROGRAM AND MODIFY.
- REAL-TIME ROBOTIC CONTROL.
- USEFUL FOR ASSISTIVE TECHNOLOGY RESEARCH.

10. FUTURE SCOPE

- INTEGRATE ARTIFICIAL INTELLIGENCE FOR IMPROVED SIGNAL CLASSIFICATION.

- INCREASE THE NUMBER OF ROBOTIC FINGER MOVEMENTS.

- DEVELOP A WIRELESS EEG ACQUISITION SYSTEM.

- IMPROVE ACCURACY USING ADVANCED MACHINE LEARNING ALGORITHMS.

- ENABLE GESTURE RECOGNITION FOR MULTIPLE HAND FUNCTIONS.

- USE MORE SOPHISTICATED BRAIN-COMPUTER INTERFACE TECHNIQUES.

CONCLUSION

THE MIND-CONTROLLED ROBOTIC HAND DEMONSTRATES THE PRACTICAL IMPLEMENTATION OF BRAIN-COMPUTER INTERFACE TECHNOLOGY USING EEG SIGNALS, THE BIO AMP EXG PILL, ARDUINO, AND SERVO MOTORS. THE PROJECT SUCCESSFULLY CONVERTS BRAIN ACTIVITY INTO ROBOTIC HAND MOVEMENT, HIGHLIGHTING THE POTENTIAL OF EEG-BASED ASSISTIVE SYSTEMS. WITH FURTHER IMPROVEMENTS IN SIGNAL PROCESSING AND ARTIFICIAL INTELLIGENCE, SUCH SYSTEMS CAN BECOME VALUABLE TOOLS FOR INDIVIDUALS WITH PHYSICAL DISABILITIES AND FUTURE HEALTHCARE APPLICATIONS.

THANK YOU!!